GHENA HANI

AI Engineer & Full-Stack Systems Architect | Deep Learning - NLP - RAG - Agentic AI

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EXECUTIVE SUMMARY

Applied AI Engineer & M.Sc. Candidate in Artificial Intelligence at USTHB (Expected June 2027) with 5 AI research internships at CERIST (2024 - Present) and 2+ years of commercial full-stack software delivery (Jan 2024 - Present). Proven track record designing evaluated deep learning architectures (MuCAT 6-language Transformer, MIT-BIH v7 INT8 TFLite ECG classifier), real-time IoT clinical platforms (MedGuardAI), hybrid RAG pipelines, and 6 deployed commercial client systems.

PROFESSIONAL & RESEARCH EXPERIENCE

AI Engineering Research Intern (5 Appointments, UbiSys Team DTISI) - CERIST, Algiers (2024 - Present)

- Designed MuCAT (Multilingual Uncertainty-Calibrated Attention Transformer) in CERIST UbiSys (DTISI): mDeBERTa-v3-base + Hierarchical Attention Pooling (HAP) + FiLM Language Gating + Evidential Dirichlet Head (u=K/S) across 6 languages (MSA, Algerian Darija, Kabyle, Chaoui, FR, EN), reaching 98.20% test accuracy vs 23.74% mDeBERTa-v3 ablation.
- Engineered AI ECG Arrhythmia Classification (v7) on the strict inter-patient MIT-BIH DS1/DS2 benchmark (90.13% 5-fold accuracy, 155.4 KB INT8 TFLite, 0.32 ms/beat latency).
- Developed real-time ECG and blood glucose anomaly detection pipelines connected to physical medical sensors and emergency clinical alert workflows.

Independent Full-Stack & Applied AI Engineer - Algiers, Algeria (Jan 2024 - Present)

- Architected and delivered 6 commercial production systems: Tadjmeel Clinica Algiers (Public Web + Doctor & Receptionist Electron Desktop Apps), Enterprise DZD Business ERP & G50 Fiscal Suite, ALLURE HOMME, GK STORE, CASUAL 29, and AURA Retail OS.
- Built and deployed 11+ interactive AI web applications spanning Speech-to-Text ASR & Clinical SOAP Summarization, Hybrid Dense+BM25 RAG, Multi-Agent ReAct Orchestration, and LLM Safety Evaluation.

KEY DEPLOYED SYSTEMS & RESEARCH

- AI ECG Arrhythmia Classification (v7): 1D-CNN + 2x Multi-Head Transformer + 8 RR features on MIT-BIH DS1/DS2.
- MedGuardAI (USTHB Capstone Distinction): Real-time IoT ECG/Glucose sensor streaming, AI anomaly alerts & WebRTC telemedicine.
- MuCAT Multilingual Transformer: mDeBERTa-v3 + HAP + FiLM Gating + Dirichlet Uncertainty (98.20% Test Acc, 6 languages).
- Commercial Production Suite: Tadjmeel Clinica Algiers, Enterprise DZD G50 ERP, ALLURE HOMME, GK STORE & CASUAL 29.

EDUCATION

M.Sc. Candidate in Artificial Intelligence - USTHB, Algiers (Sep 2025 - June 2027 Expected)

B.Sc. (Licence) in Computer Science - USTHB, Algiers (Sep 2022 - June 2025 Graduated - Capstone: MedGuardAI)

TECHNICAL COMPETENCIES & LANGUAGES

AI & Deep Learning: PyTorch, TensorFlow, Keras, Transformers, BERT, RAG, Agentic AI (ReAct / Tool Calling), LLM Evaluation, TFLite INT8, Computer Vision.

Full-Stack & Cloud: TypeScript, React, Next.js, Node.js, Electron, PostgreSQL, MongoDB, Supabase, REST/WebSockets, Docker, Git.